MATRIX CHAIN OPTIMIZER 3D

Interval Dynamic Programming Visualizer

Team 09

Project Description

Matrix Chain Optimizer 3D is a Java-based web application that demonstrates how interval dynamic programming finds an efficient order for multiplying a sequence of matrices. The order of the matrices remains unchanged, but changing their parenthesization can significantly affect the number of scalar multiplications.

The application allows users to configure the number of matrices and enter compatible dimensions. It computes the minimum multiplication cost, derives the optimal parenthesization, compares the result with left-to-right multiplication, and displays the dynamic programming cost table.

Key Features

- Accepts the number of matrices and their dimensions as input.
- Builds matrix representations from the configured dimensions.
- Calculates minimum scalar multiplication cost using interval dynamic programming.
- Displays the optimal matrix multiplication order (parenthesization).
- Compares optimized cost with naive left-to-right multiplication.
- Reports operations saved and percentage efficiency improvement.
- Displays the DP cost table for matrix intervals.
- Provides a 3D-style visual representation of the matrix chain.
- Includes Play, Pause, Previous, Next, and Reset controls for the visualization.
- Provides a responsive browser interface for learning the optimization process.

Technologies Used

- Java 17
- Apache Maven
- HTML
- CSS
- JavaScript
- Java built-in HTTP server
- Interval Dynamic Programming

Algorithm

The project uses the Matrix Chain Multiplication recurrence:

```
dp[i][j] = min over i ≤ k < j of { dp[i][k] + dp[k+1][j] + p[i−1] × p[k] × p[j] }
```

Time Complexity: $O(n^3)$
Space Complexity: $O(n^2)$

Expected Outcome

The application demonstrates how dynamic programming can reduce unnecessary scalar multiplications by choosing an efficient parenthesization. It presents numerical cost comparisons, the optimal order, a DP table, and an interactive visual representation to make the optimization process easier to understand.